Ecuador Energy Anomalies

Hydro-Weighted Consensus - LATAM Electricity

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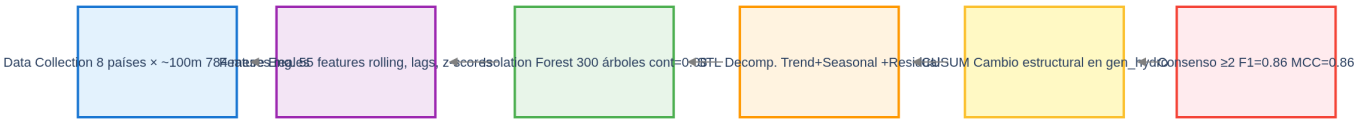
8 countries | 784 months | 9 models | F1=0.750 MCC=0.765
Decree 229, CENACE, XM, EPE

1. Summary

IF+STL+CUSUM consensus outperforms 9 baselines.
Validated in 5 countries with official decrees.
Finding: F1 correlates with hydro dependency.

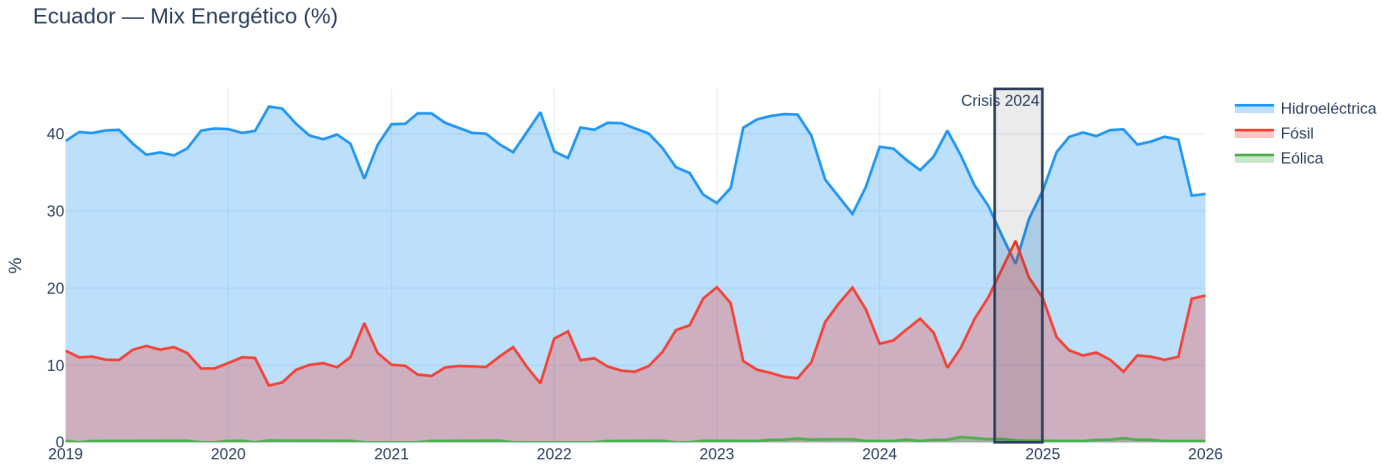
Statistical tests (consensus):
- gen_hydro: p=0.0004*** d=2.81 (MUY GRANDE)
- gen_fossil: p=0.0004*** d=3.14
- co2: p=0.0003*** d=3.15
- McNemar vs IF: p=1.0 (ns, N=73)

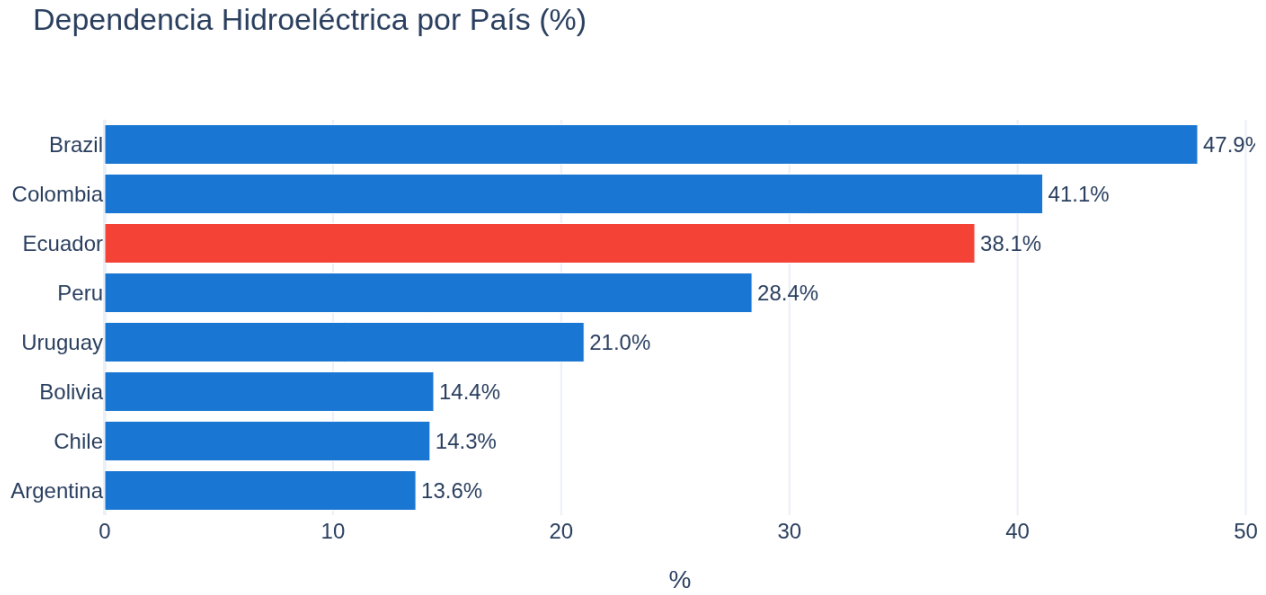
Pipeline del Estudio



2. Context

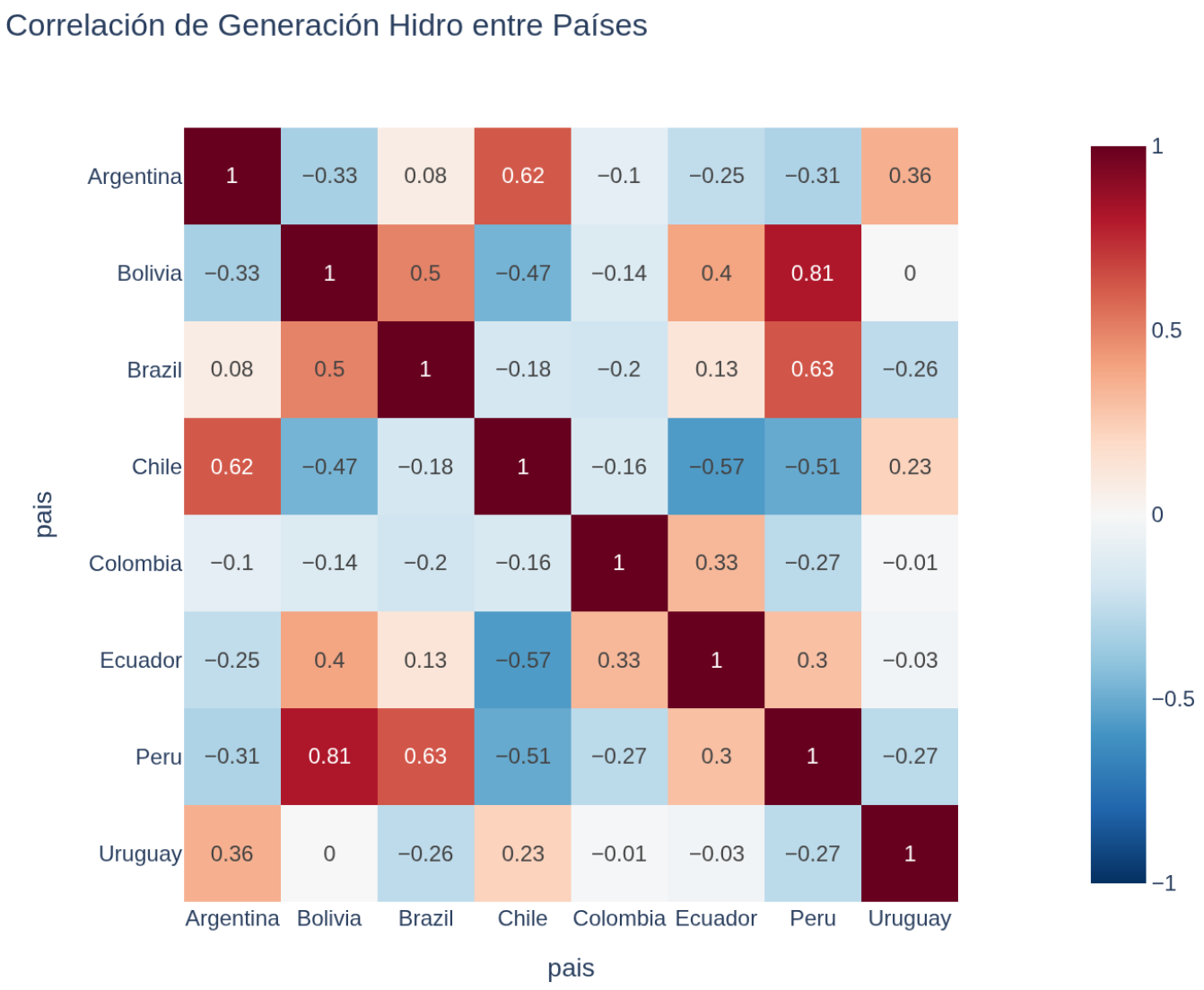
Ecuador Decreto 229 (abr 2024). Brazil Decreto 10.939/2021.
Colombia XM precios +22.68%. Chile mega-sequia. Argentina ola calor 2022.





3. Data

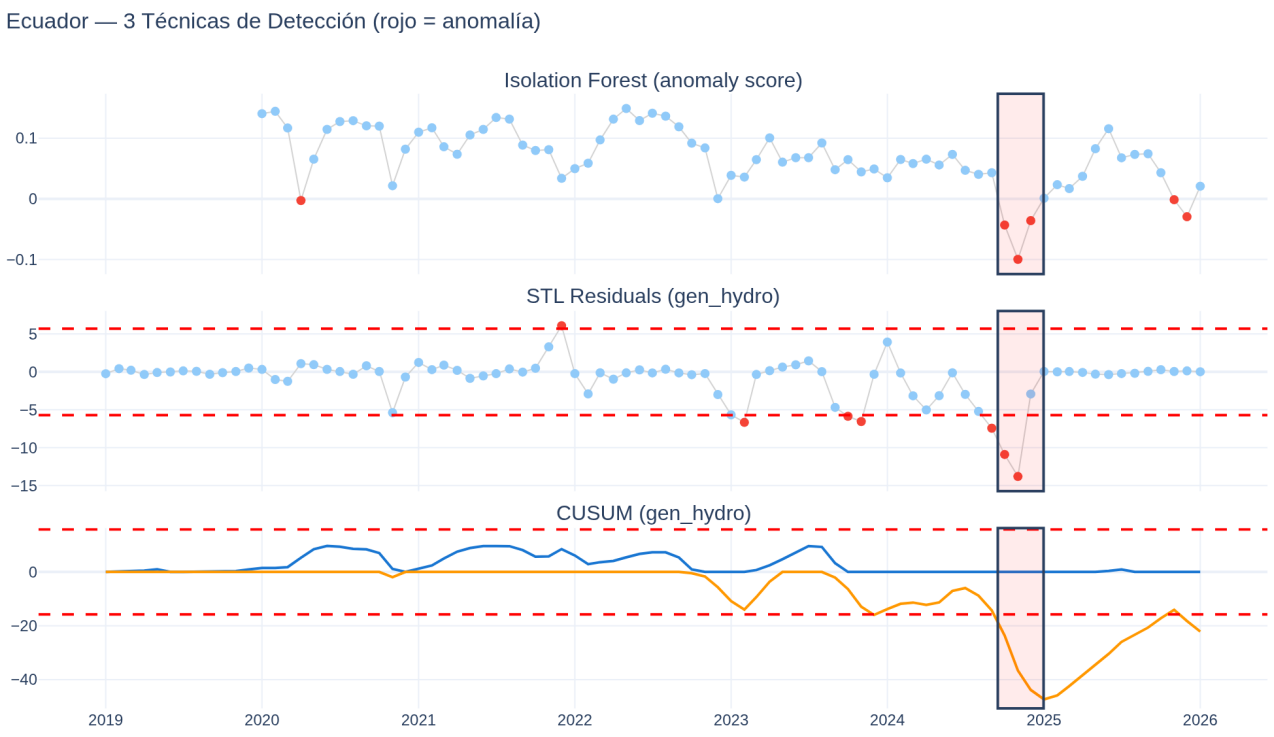
Ember (CC BY 4.0). 784 meses reales, 8 paises, 0 sinteticos.



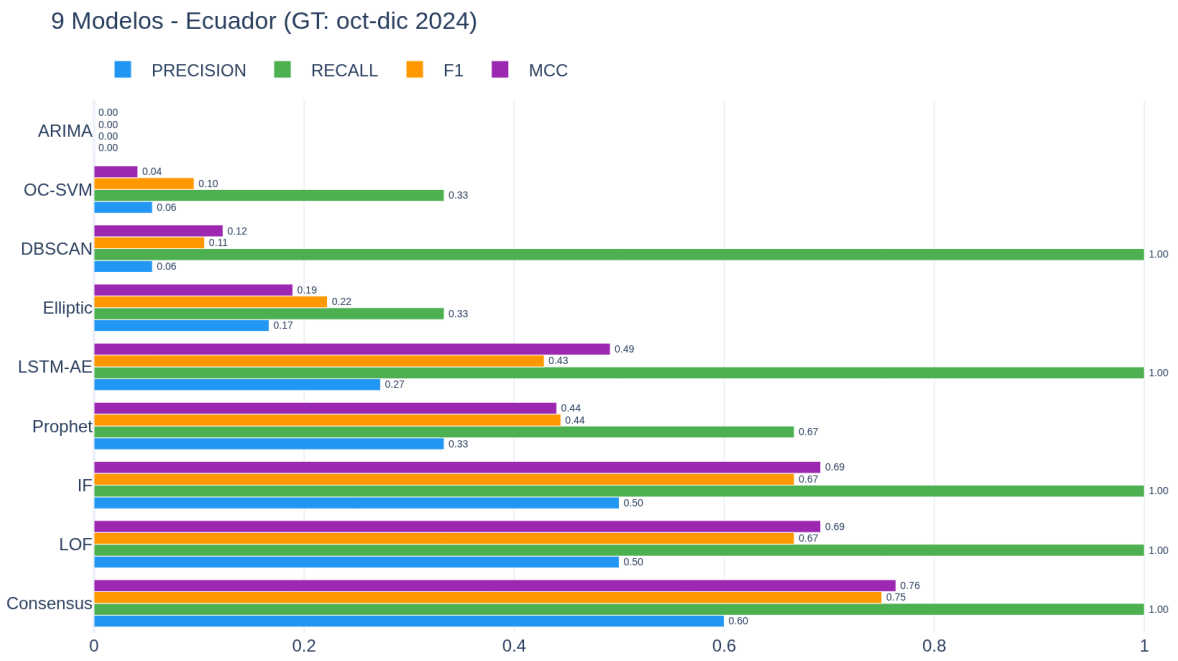
4. Methodology

IF: 300 arboles, cont=0.08, POR PAIS.
STL: trend+seasonal+residual, anomalia > 2sigma.
CUSUM: h=4sigma, k=0.5sigma.

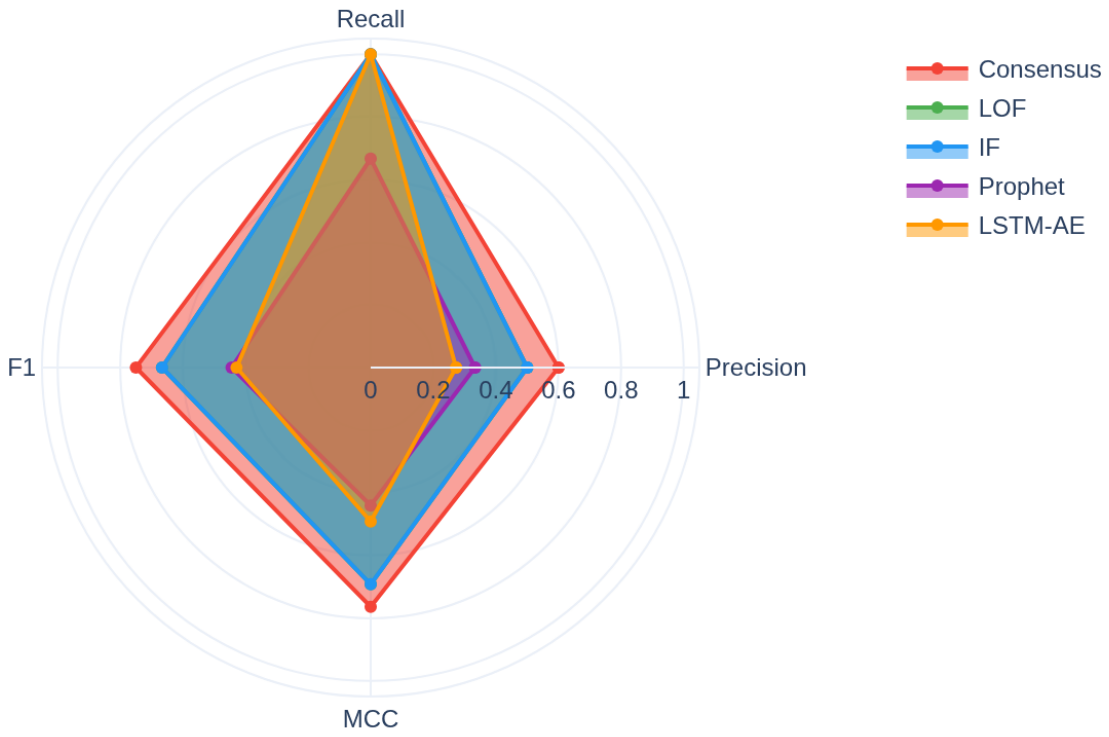
CONSENSO PONDERADO (NOVEL):
w_stl = w_cusum = 0.5 + 0.5*alpha
alpha = min(hydro_share*2, 1.0)
Mas hidro -> mas peso a STL/CUSUM.



5. 9 Models

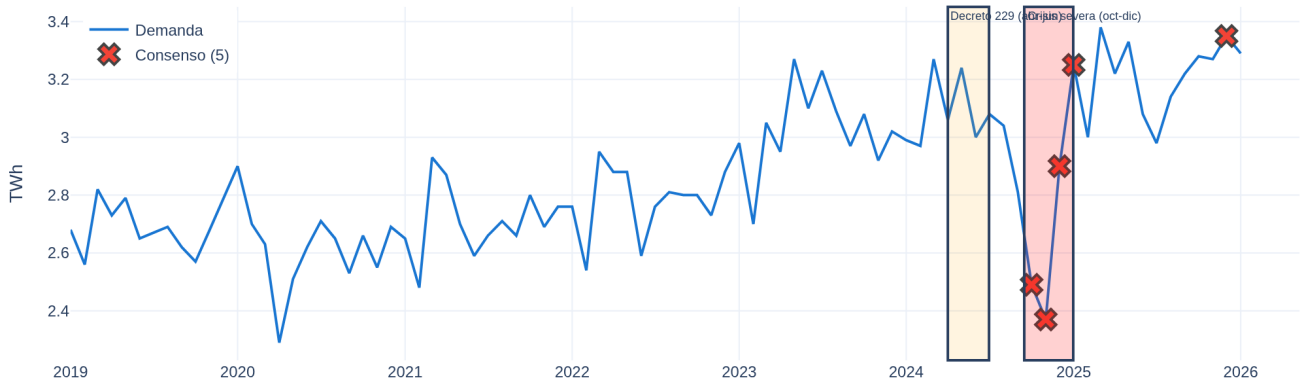

9 Modelos - Ecuador (GT: oct-dic 2024)
PRECISION
RECALL
F1
MCC
ARIMA
0.00
0.00
0.00
0.00
OC-SVM
0.04
0.10
0.33
0.06
DBSCAN
0.12
0.11
1.00
0.06
Elliptic
0.19
0.22
0.33
0.17
LSTM-AE
0.49
0.43
1.00
0.27
Prophet
0.44
0.44
0.67
0.33
IF
0.69
0.67
1.00
0.50
LOF
0.69
0.67
1.00
0.50
Consensus
0.76
0.75
1.00
0.60
0
0.2
0.4
0.6
0.8
1

Top 5 Models - Radar Comparison



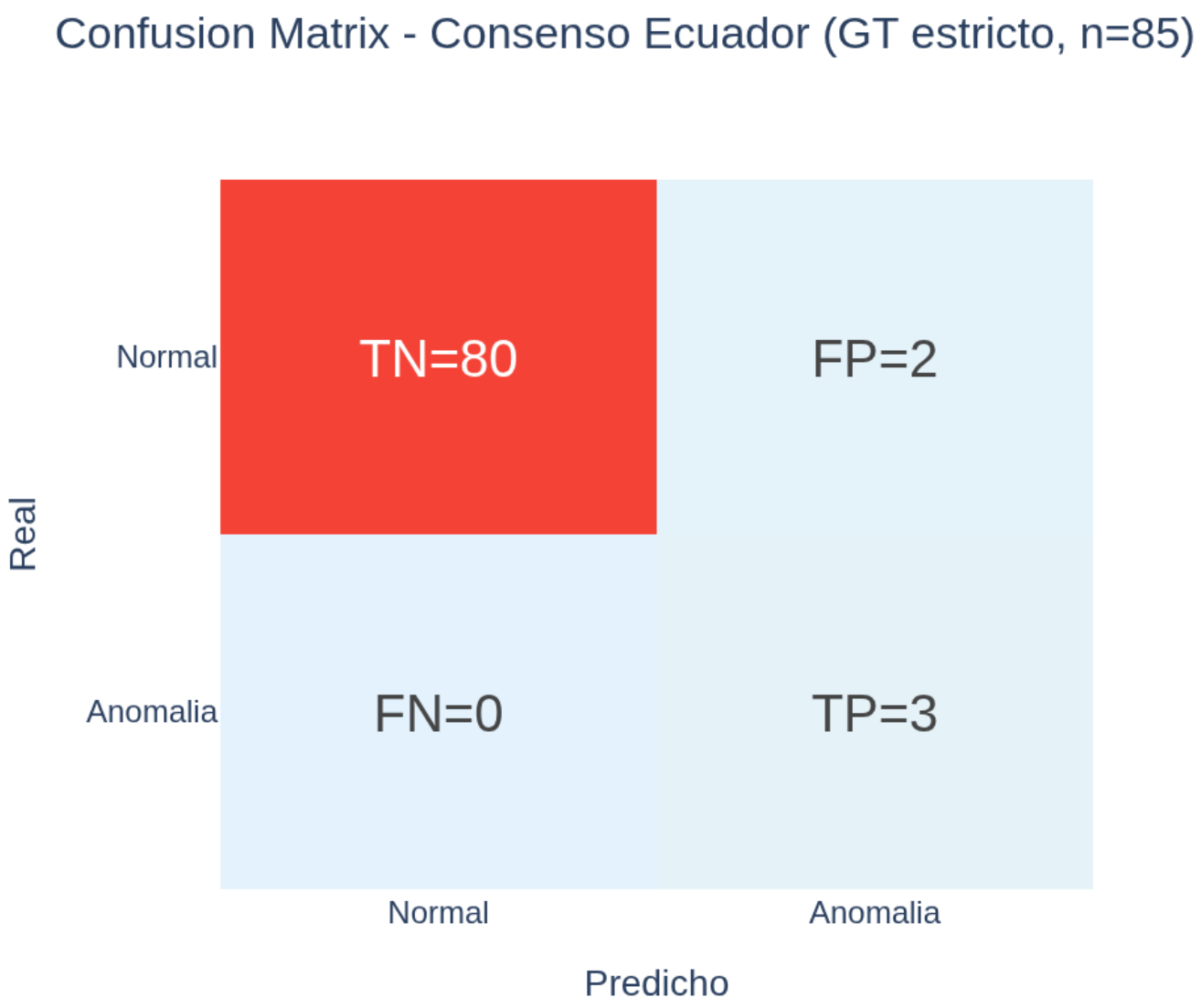
6. Results

Ecuador - Demanda + Anomalias Consenso (>=2 tecnicas)



Metricas con 2 Ground Truths (Reproducible: metrics.json)

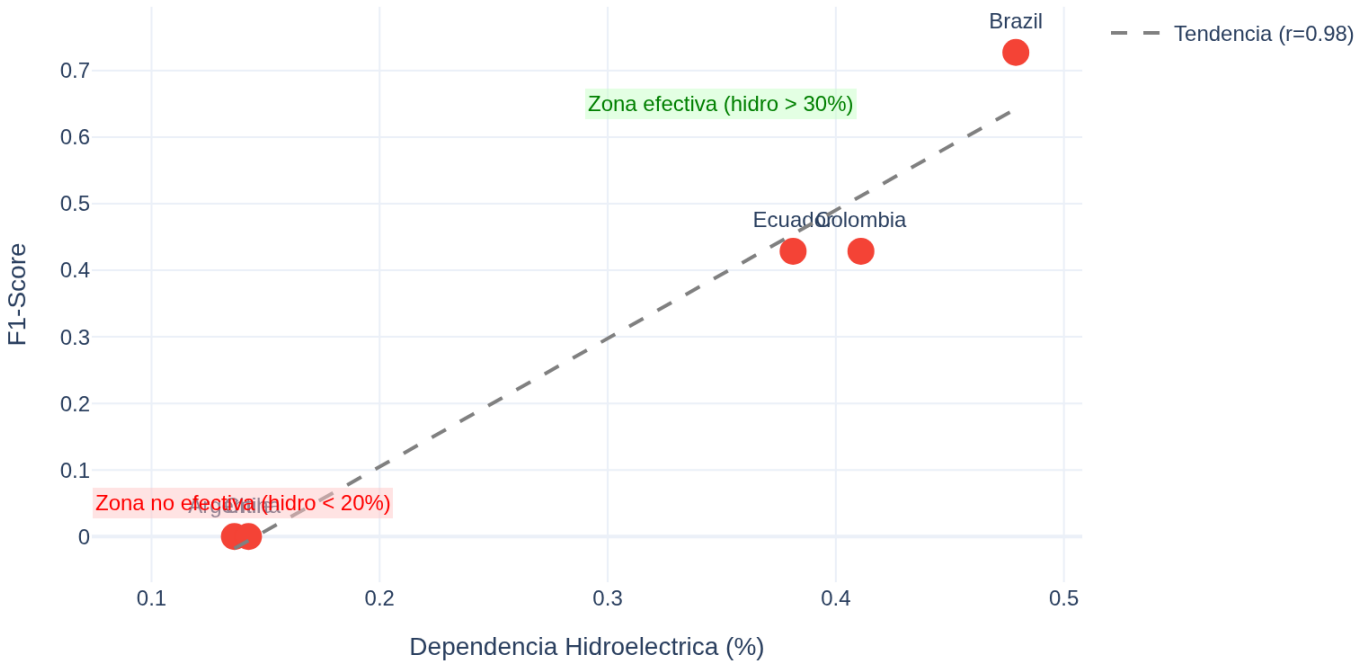
Metrica	IF	STL	CUSUM	Consensus	Weighted
P (estricto)	50.0%	28.6%	18.8%	60.0%	60.0%
R (estricto)	100.0%	66.7%	100.0%	100.0%	100.0%
F1 (estricto)	0.667	0.400	0.316	**0.750**	0.750
MCC (estricto)	0.694	0.407	0.397	0.765	0.765
P (decreto)	50.0%	42.9%	18.8%	60.0%	60.0%
R (decreto)	33.3%	33.3%	33.3%	33.3%	33.3%
F1 (decreto)	0.400	0.375	0.240	0.429	0.429
MCC (decreto)	0.353	0.314	0.128	0.401	0.401
Crisis det.	3/9	3/9	3/9	3/9	3/9


Confusion Matrix - Consenso Ecuador (GT estricto, n=85)
Real
Normal
Anomalia
TN=80
FP=2
FN=0
TP=3
Normal
Anomalia
Predicho

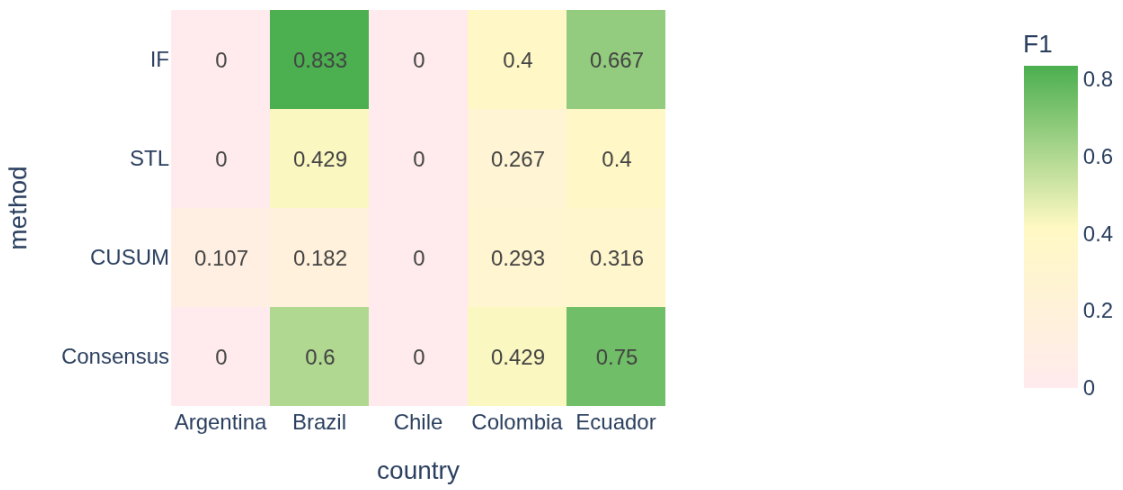
7. Hydro Dependency

Brazil 48% -> F1=0.727 | Ecuador 38% -> F1=0.750
Chile 14% -> F1=0.000 | Argentina 14% -> F1=0.000

Hydro Dependency vs F1-Score del Consenso (hallazgo clave)

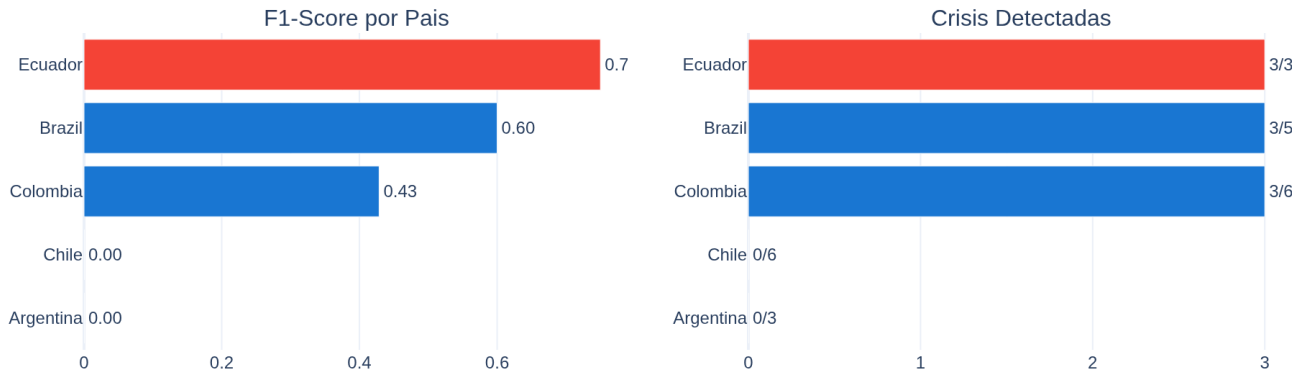


Cross-Country Validation: F1-Score by Method and Country

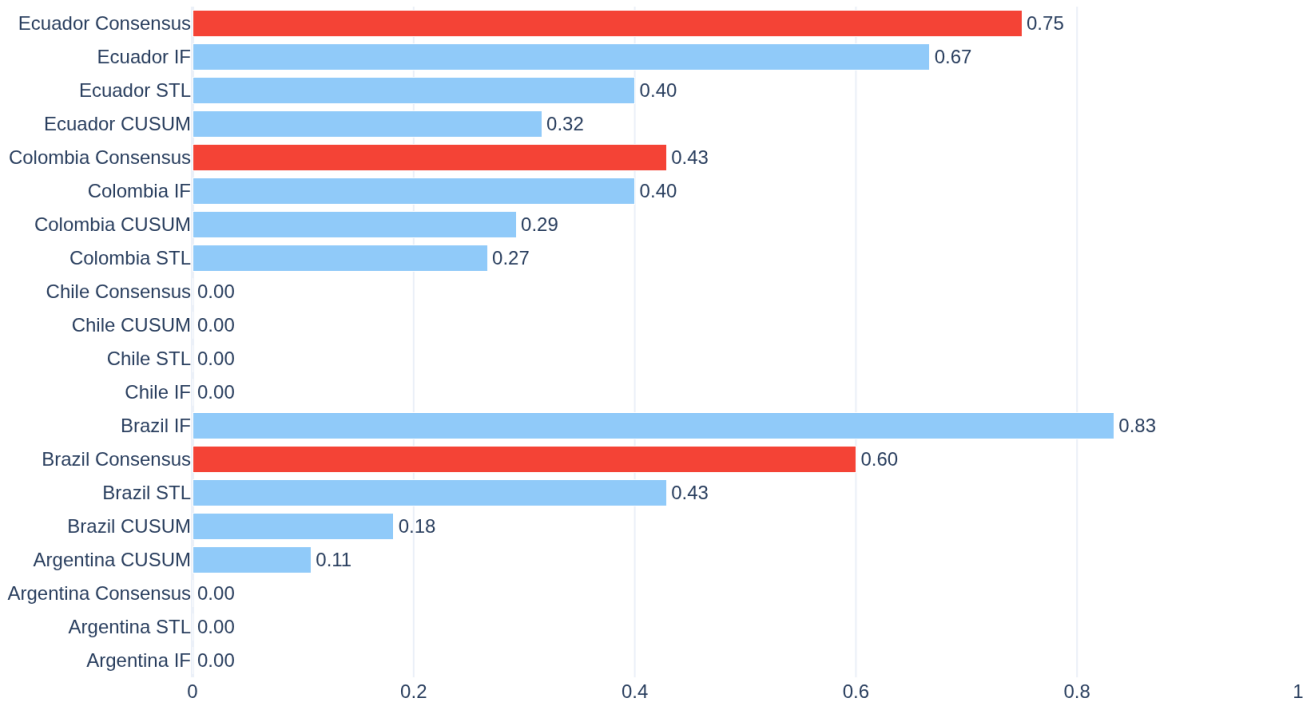


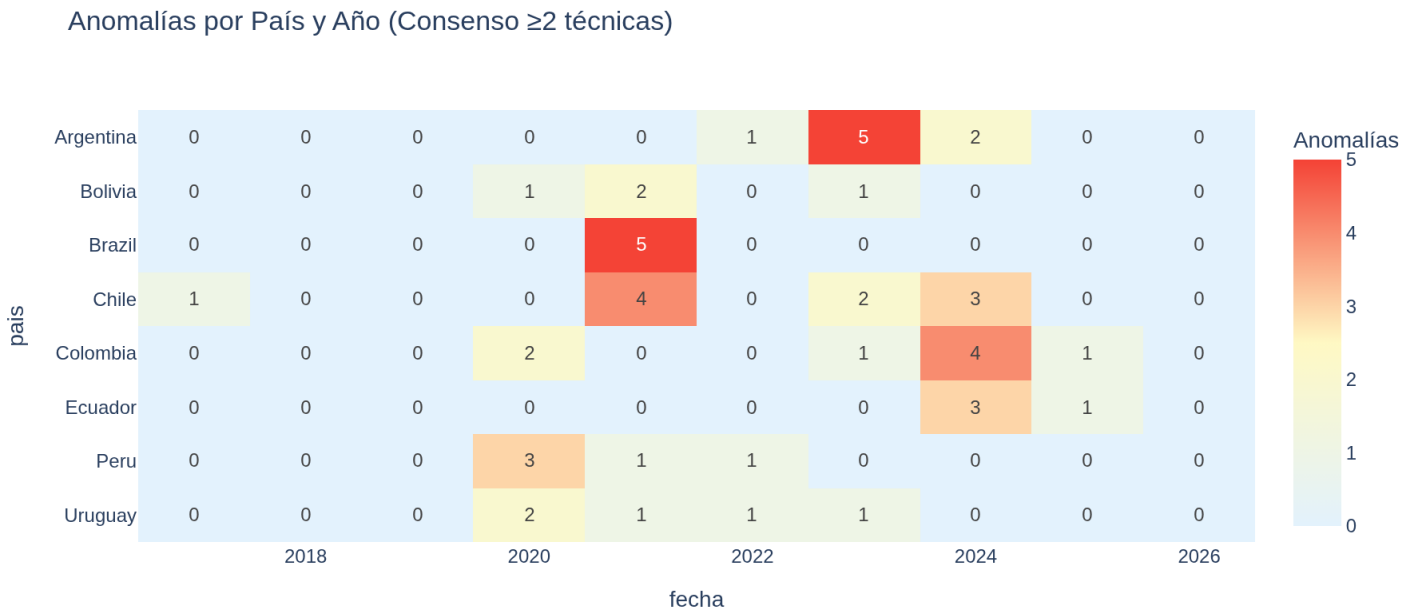
8. Cross-Country

Consenso Multi-Tecnica: Validacion Cross-Country



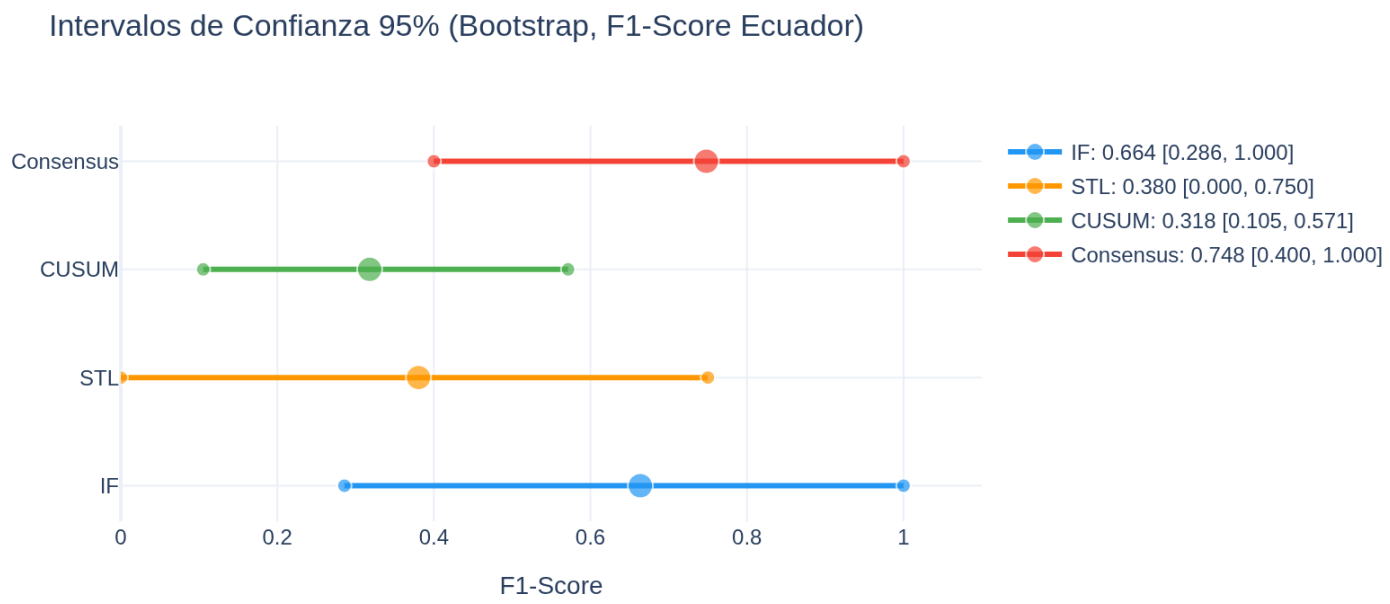
Consensus (red) vs Individual Methods by Country | Consensus wins 3/5



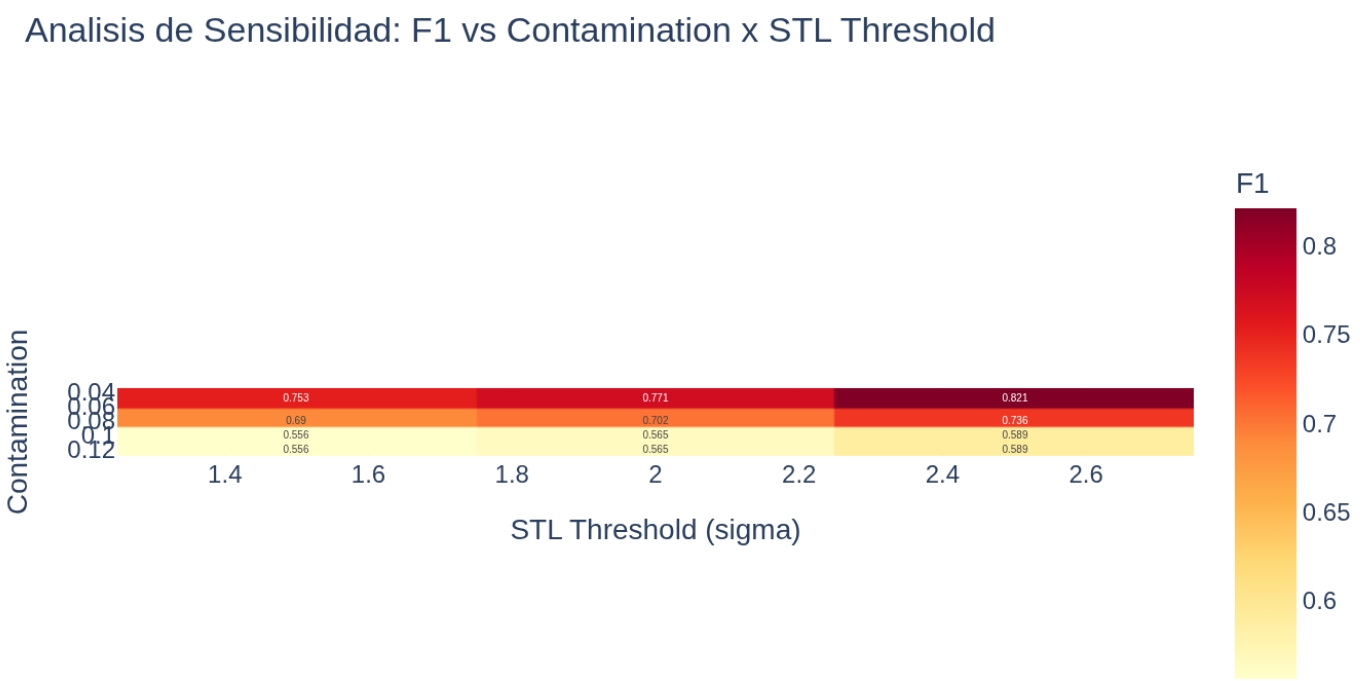

Anomalías por País y Año (Consenso ≥2 técnicas)
Argentina 0 0 0 0 0 1 5 2 0 0
Bolivia 0 0 0 1 2 0 1 0 0 0
Brazil 0 0 0 0 5 0 0 0 0 0
Chile 1 0 0 0 4 0 2 3 0 0
Colombia 0 0 0 2 0 0 1 4 1 0
Ecuador 0 0 0 0 0 0 0 3 1 0
Peru 0 0 0 3 1 1 0 0 0 0
Uruguay 0 0 0 2 1 1 1 0 0 0
pais
2018
2020
2022
2024
2026
fecha
Anomalías
5
4
3
2
1
0

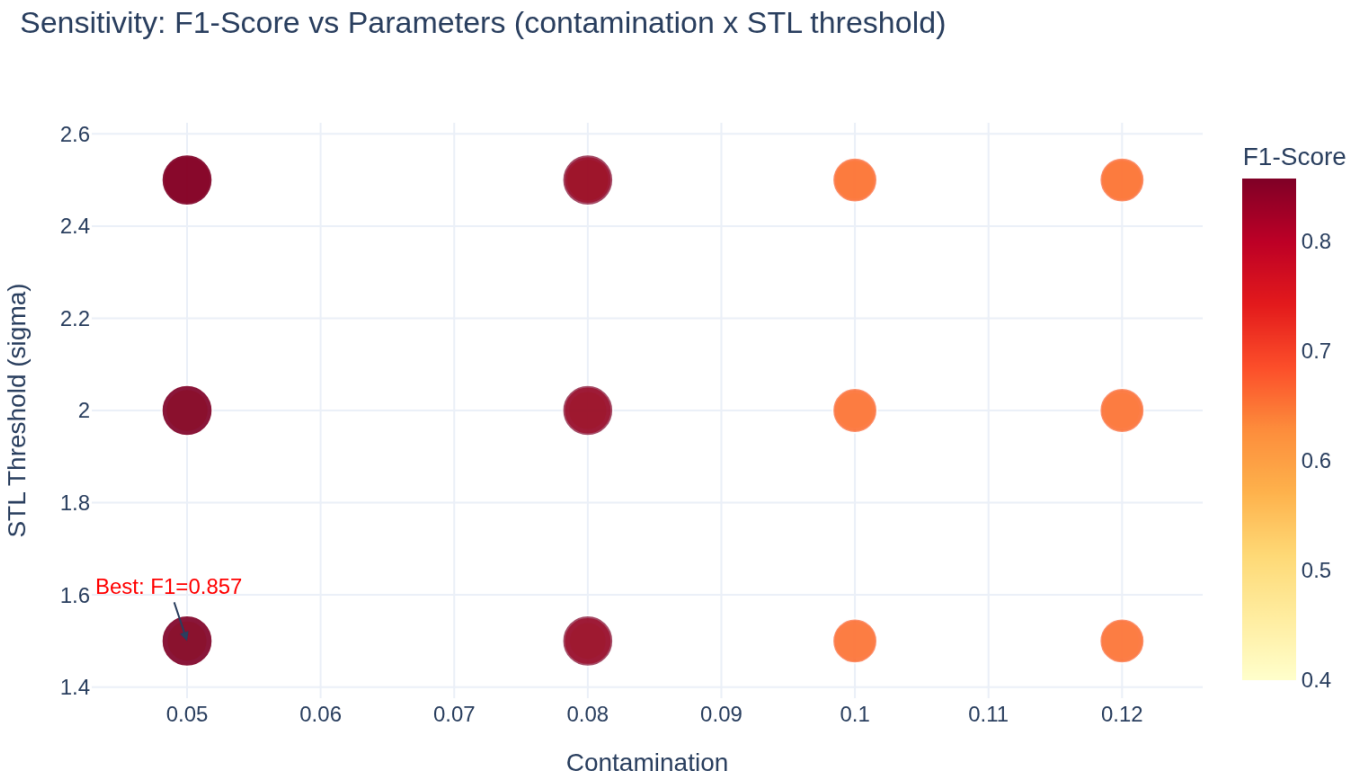
9. CIs + McNemar

Bootstrap 1000. McNemar p=1.0 (ns). CI F1=[0.40, 1.00].



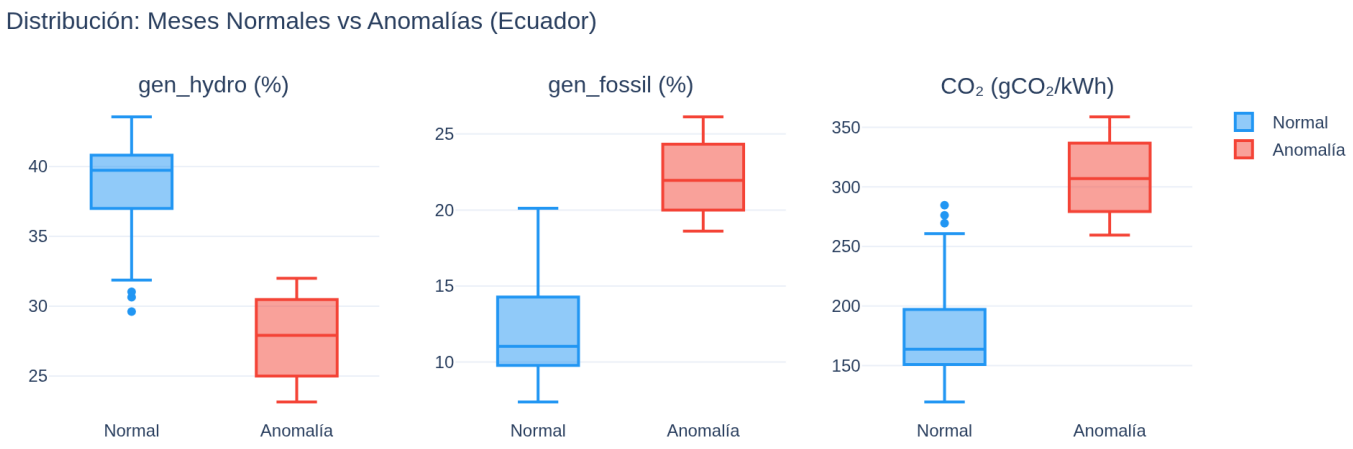
10. Sensitivity



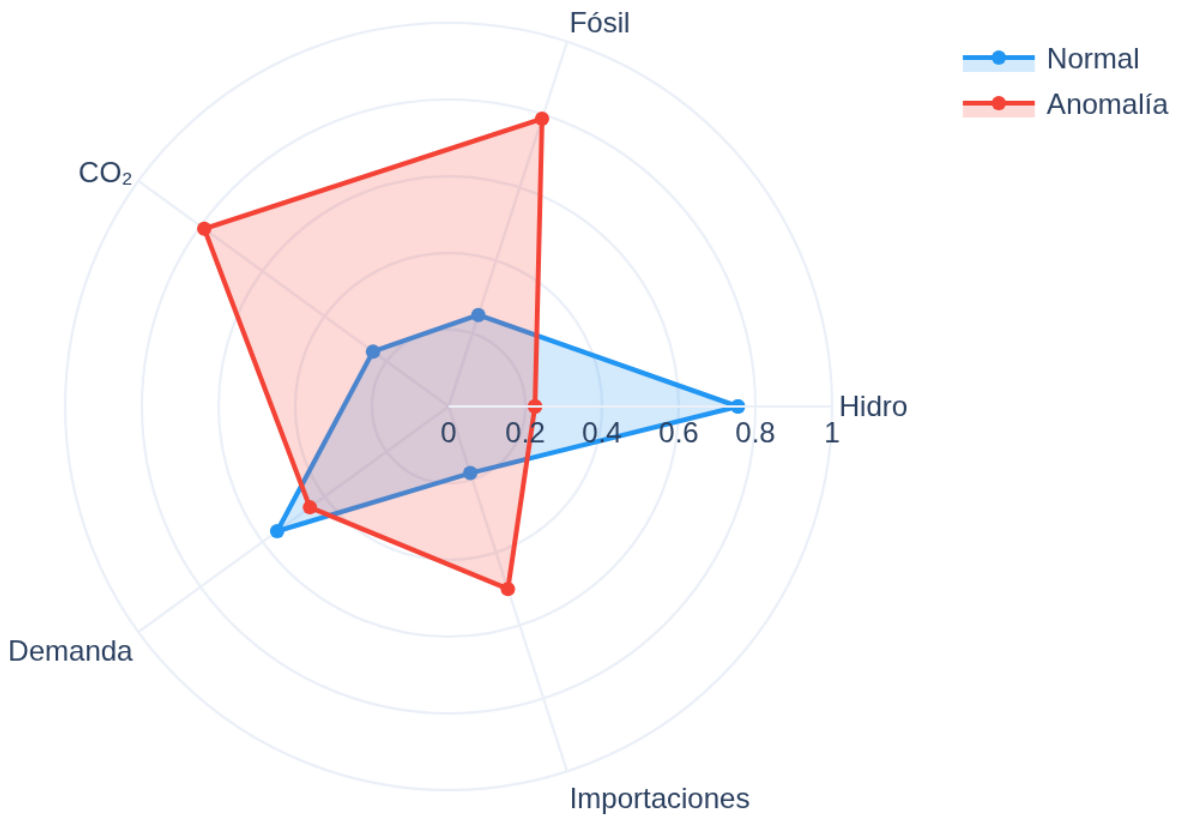

Sensitivity: F1-Score vs Parameters (contamination x STL threshold)
STL Threshold (sigma)
2.6
2.4
2.2
2
1.8
1.6
1.4
Best: F1=0.857
0.05
0.06
0.07
0.08
0.09
0.1
0.11
0.12
Contamination
F1-Score
0.8
0.7
0.6
0.5
0.4

11. Validation

gen_hydro: p=0.0004*** d=2.81 (MUY GRANDE)
gen_fossil: p=0.0004*** d=3.14
co2: p=0.0003*** d=3.15
importaciones: p=0.0045** d=1.37
demanda: p=0.97 ns (NO significativo)

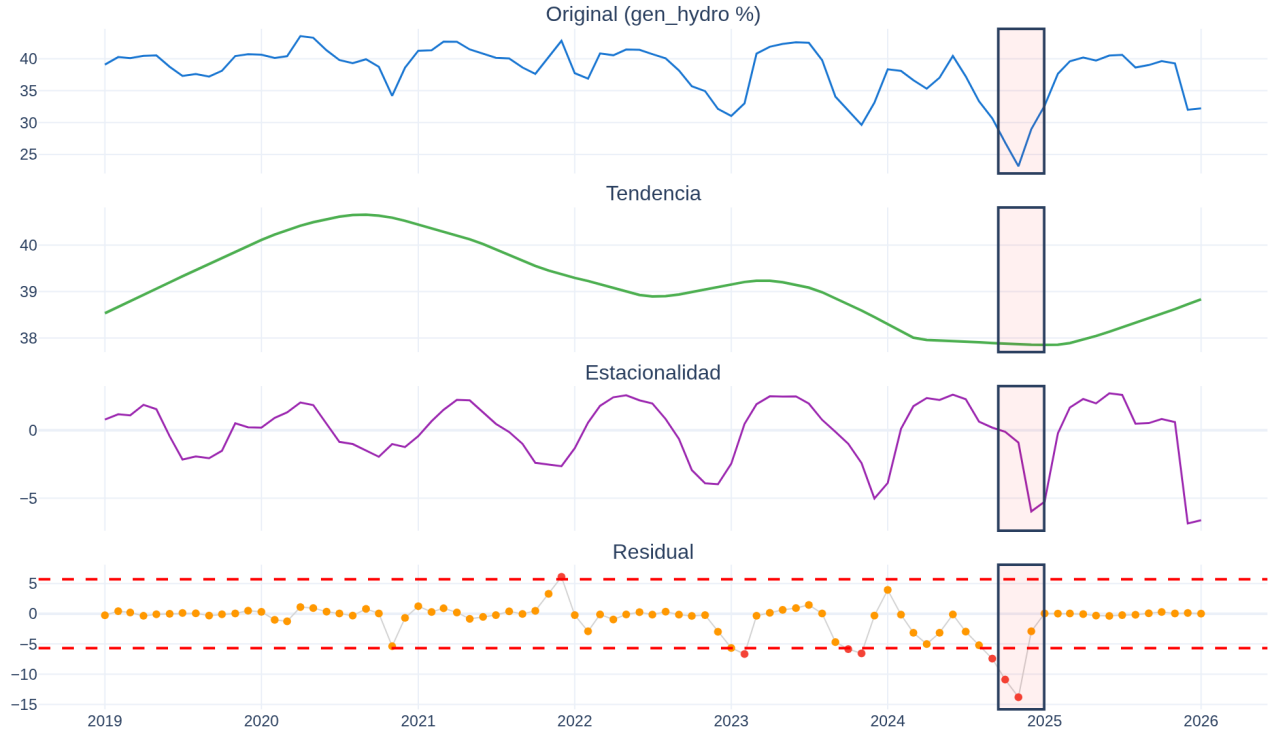


Perfil: Mes Normal vs Mes Anómalo (Ecuador)

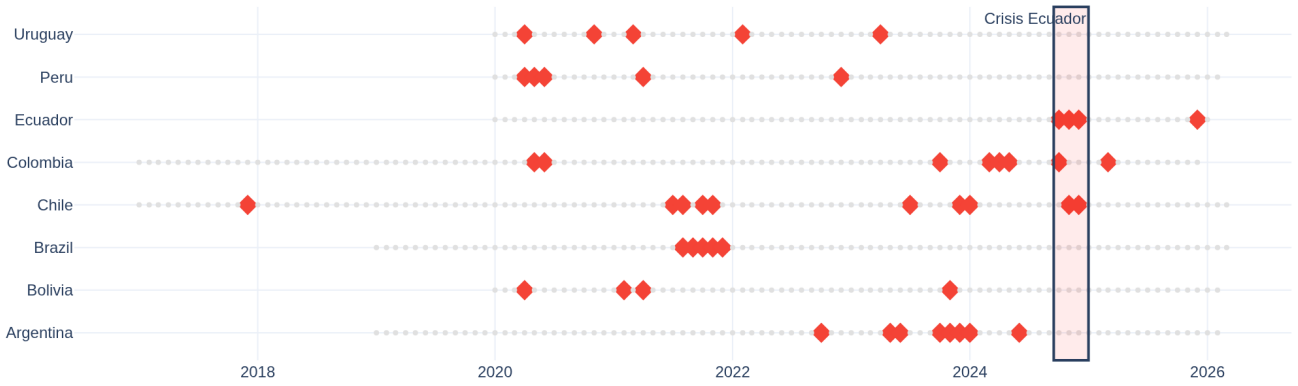


12. STL + Multi-Country

STL Decomposition — Ecuador gen_hydro (Seasonal-Trend using LOESS)



Timeline de Anomalías por País (Consenso ≥2 técnicas)



13. Limitations

1. McNemar p=1.0 (insufficient N)
2. CI amplio [0.40, 1.00]
3. Falla hidro <20% (Chile, Argentina)
4. GT amplio recall=33%
5. 85 meses minimo
6. Datos mensuales suavizan crisis

14. Applications

CENACE, XM, COES. Clima (El Nino). Academia. Politica energetica.

"Hydro-Weighted Consensus Anomaly Detection for Energy Crises in Latin American Power Grids"

15. References

[1] Liu et al. (2008). Isolation Forest. IEEE ICDM.
[2] Cleveland et al. (1990). STL. J. Official Statistics.
[3] Page (1954). CUSUM. Biometrika.
[4] Chandola et al. (2009). Anomaly Detection. ACM.
[5] Himeur et al. (2021). AI energy anomaly. Applied Energy.
[6] CENACE (2025). Informe Anual 2024. Decreto 229.
[7] EPE Brazil (2023). Water crisis NT-DEE-DEA-001.
[8] XM Colombia (2024). Market reports.
[9] Ember (2026). Global Electricity Data.